

NLP

- What is Natural Language Processing?
 - Applications of NLP:
 - Text Classification (Spam Detection, Sentiment Analysis)
 - Named Entity Recognition (NER)
 - Machine Translation
 - Question Answering Systems
 - Chatbots and Conversational Agents
 - Text Summarization
 - Overview of NLP Libraries:
 - NLTK, SpaCy, Gensim, Hugging Face Transformers, TextBlob
 - Text Preprocessing in NLP
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2. Mathematical Foundations

2.1 Probability and Statistics for NLP

- Probability Theory (Bayes Theorem, Conditional Probability)
- Random Variables and Distributions
- Expectation and Variance
- Markov Chains and Hidden Markov Models (HMM)
- Maximum Likelihood Estimation (MLE)

2.2 Linear Algebra

- Vectors, Matrices, and Tensors
- Singular Value Decomposition (SVD)
- Eigenvalues and Eigenvectors
- Principal Component Analysis (PCA)

2.3 Information Retrieval and Theory

- Entropy, Cross-Entropy, and KL-Divergence
 - TF-IDF (Term Frequency - Inverse Document Frequency)
 - Word Embeddings and Vector Spaces
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3. Text Preprocessing and Representation

3.1 Text Tokenization

- Tokenization: Sentence vs. Word Tokenization
- Regular Expressions for Tokenization
- WordPunctTokenizer, TreebankWordTokenizer

3.2 Removing Noise and Text Cleaning

- Lowercasing, Punctuation Removal
- Stopwords Removal
- Stemming vs. Lemmatization
 - Porter Stemmer, Lancaster Stemmer
 - WordNet Lemmatizer

3.3 Vectorizing Text Data

- Bag of Words (BoW)
- TF-IDF Vectorizer
- One-Hot Encoding
- CountVectorizer and TfidfVectorizer in scikit-learn

3.4 Word Embeddings

- Introduction to Word Embeddings
- Word2Vec (Skip-gram and CBOW)
- GloVe (Global Vectors for Word Representation)
- FastText

3.5 Advanced Text Representation

- Distributed Representations of Text
 - Contextual Word Embeddings (ELMo, BERT, GPT)
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4. Syntax and Parsing

4.1 Part-of-Speech (POS) Tagging

- POS Tagging and Its Importance
- POS Tagging Algorithms (Rule-based, HMM)
- Using SpaCy for POS Tagging

4.2 Named Entity Recognition (NER)

- What is NER?
- NER Techniques: Rule-based, Statistical, Deep Learning
- Using SpaCy for Named Entity Recognition

4.3 Parsing Techniques

- Dependency Parsing vs. Constituency Parsing
- Parse Trees and Syntax Trees
- Using SpaCy for Dependency Parsing

4.4 Sentence Structure and Syntactic Relations

- Constituency Grammar
 - Dependency Grammar
 - Parsing Algorithms (Earley, CYK)
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5. Text Classification and Feature Extraction

5.1 Text Classification

- Supervised vs. Unsupervised Classification
- Feature Extraction Methods (BoW, TF-IDF)
- Classification Algorithms for Text (Logistic Regression, Naive Bayes, SVM)
- Evaluating Models: Precision, Recall, F1-Score, ROC-AUC

5.2 Advanced Text Classification Techniques

- Multiclass vs. Multilabel Classification
- Ensemble Methods: Random Forest, XGBoost
- Neural Networks for Text Classification (Feedforward Networks, CNNs, LSTMs)
- Handling Imbalanced Data in Text Classification

5.3 Document Similarity and Clustering

- Cosine Similarity and Euclidean Distance
 - K-Means Clustering for Text
 - Hierarchical Clustering
 - DBSCAN for Document Clustering
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6. Named Entity Recognition (NER)

6.1 Introduction to NER

- What is Named Entity Recognition?
- Named Entities: People, Locations, Organizations, Dates, etc.
- NER Algorithms: Rule-based, Statistical Models, and Deep Learning Approaches

6.2 Deep Learning for NER

- BiLSTM (Bidirectional Long Short-Term Memory) for Sequence Labeling
 - CRF (Conditional Random Fields) for NER
 - Fine-tuning Pre-trained BERT for NER
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7. Language Modeling

7.1 Introduction to Language Models

- N-grams and Markov Chains for Language Modeling
- Probability of Word Sequences and Likelihood Estimation
- Smoothing Techniques (Additive, Kneser-Ney)

7.2 Neural Language Models

- Introduction to RNNs (Recurrent Neural Networks)
 - LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Units)
 - Transformer Models for Language Modeling
 - Training and Fine-tuning Large Pretrained Language Models (GPT-2, GPT-3, BERT, T5)
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8. Sequence Labeling and Tagging

8.1 Sequence Labeling Tasks

- POS Tagging, NER, Chunking
- Hidden Markov Models for Sequence Labeling
- CRFs (Conditional Random Fields) for Sequence Labeling

8.2 Recurrent Neural Networks (RNN) for Sequence Labeling

- RNN Basics: Vanishing and Exploding Gradients
 - LSTMs and GRUs for Sequence Labeling
 - Bidirectional LSTMs for Contextual Information
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9. Text Generation

9.1 Introduction to Text Generation

- Basics of Text Generation Models
- Markov Chains for Simple Text Generation

9.2 Recurrent Neural Networks for Text Generation

- Generating Text with LSTMs
- Training LSTMs on Character-level and Word-level Data

9.3 Transformer-based Text Generation

- Generative Pre-trained Transformers (GPT-2, GPT-3)
 - Fine-tuning GPT Models for Custom Text Generation Tasks
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10. Machine Translation

10.1 Statistical Machine Translation (SMT)

- Phrase-based Translation Models
- Word Alignment and Translation Models

10.2 Neural Machine Translation (NMT)

- Sequence-to-Sequence (Seq2Seq) Models for NMT
 - Attention Mechanism in NMT
 - Using Pretrained Translation Models (e.g., MarianMT)
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11. Question Answering Systems

11.1 Types of Question Answering

- Closed-domain vs. Open-domain Question Answering
- Extractive vs. Abstractive Question Answering

11.2 Traditional QA Systems

- Rule-based Question Answering
- Information Retrieval for QA (TF-IDF, BM25)

11.3 Deep Learning for QA

- Using BiDAF (Bi-directional Attention Flow) for QA
 - Fine-tuning BERT for Question Answering
 - SQuAD (Stanford Question Answering Dataset) and Evaluation
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12. Text Summarization

12.1 Extractive Summarization

- TextRank and LexRank Algorithms
- Using TF-IDF for Extractive Summarization
- Sentence Selection Methods

12.2 Abstractive Summarization

- Sequence-to-Sequence Models for Abstractive Summarization
 - Attention Mechanism for Summarization
 - Using Pretrained BERT and GPT for Summarization Tasks
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13. Sentiment Analysis

13.1 Introduction to Sentiment Analysis

- Analyzing Sentiments (Positive, Negative, Neutral)
- Sentiment Lexicons: AFINN, VADER, SentiWordNet
- TextBlob for Sentiment Analysis

13.2 Machine Learning for Sentiment Analysis

- Supervised Learning Approaches (Logistic Regression, SVM)
- Fine-tuning BERT for Sentiment Classification

13.3 Deep Learning for Sentiment Analysis

- Using CNNs, RNNs, and LSTMs for Sentiment Classification
 - Multi-class Sentiment Classification
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14. Advanced Topics

14.1 Transfer Learning in NLP

- Fine-tuning Pre-trained Models (BERT, GPT-3, RoBERTa)
- Using Hugging Face Transformers for Transfer Learning

14.2 Ethics and Bias in NLP

- Ethical Challenges in NLP
 - Addressing Bias in Language Models
 - Fairness in Sentiment Analysis and Text Classification
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15. NLP Projects and Case Studies

Beginner Projects

- Sentiment Analysis on Social Media Data
- Text Classification with Naive Bayes or SVM
- Build a Simple Chatbot with Rule-based Methods

Intermediate Projects

- Named Entity Recognition using SpaCy
- Machine Translation using Sequence-to-Sequence Models
- Build a Question Answering System using BERT

Advanced Projects

- Fine-tune GPT-3 for Text Generation and Creative Writing
- Build a Summarization Tool using Transformer-based Models

- Implement a Real-time Sentiment Analysis Web Application